

Massive Grant-Free Random Access in Cell-Free Massive MIMO URLLC Systems

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Abstract—Next generation wireless network is expected to provide higher rate, more reliable access and lower latency. Cell-free massive multiple input multiple output (CF-mMIMO) has been considered a potential enabler for massive access. In this paper, we primarily investigate massive grant-free random access (GFRA) in CF-mMIMO ultra-reliable and low latency (URLLC) systems. First, we give a massive GFRA model based on CF-mMIMO URLLC system, focusing on analyzing access reliability and access latency. Next, we derive approximated closed-form expression for the decoding error probability and outage probability of active users in the finite block-length regime by leveraging the approximated distribution of the signal to interference plus noise power. Based on this, by comprehensively considering the random access procedure, we derive the expression for access success probability of the attempted access user. We also jointly optimize the pilot length and data block-length by the access success probability to enhance access reliability. Furthermore, by utilizing the macro diversity of the CF-mMIMO system, we propose a scalable user centric-based multiple access points scheme to improve access reliability. In addition, based on access procedure and the associated GFRA protocol, we analyze the components of access latency. Subsequently, considering the access success probability and the number of retransmissions, we analyze and evaluate access latency for massive GFRA in the CF-mMIMO URLLC system. Finally, simulation results demonstrate the rationality and effectiveness of the proposed massive access model for analyzing the access reliability and access latency.

Index Terms—Cell-free massive MIMO, ultra-reliable and low-latency communications (URLLC), Access success probability, Grant free random access.

I. INTRODUCTION

WITH the rapid development of various Internet of Things (IoT) vertical services, such as autonomous driving vehicles, factory automation and augmented reality, the number of wireless terminals in beyond the fifth generation (B5G) and sixth generation (6G) wireless networks is also increasing rapidly. Meanwhile, stringent latency requirements and access reliability have become crucial [1]. Therefore, the evolution of future communications networks supporting massive machine-type communication (mMTC) and ultra-reliable and low-latency communications (URLLC) needs to

provide greater system capacity, higher rates, lower latency, and more reliable access [2]. URLLC is a critical enabler of emerging applications such as industrial automation and factory control, autonomous driving, tactile Internet, and aerial and drone communications [3]. These use scenarios typically involve short, sporadic, and latency-critical transmissions that must be delivered with high reliability and minimal delay. To achieve high-reliability and low-latency communication for massive access at any time and anywhere, research on massive access technologies that satisfy URLLC demands in mMTC scenarios has become a research hot spot [4].

URLLC and mMTC are two of the three core businesses defined in the 5G specification. URLLC aims to achieve less than 1 ms user plane latency for 32-byte packet transmission with greater than $1 - 10^{-5}$ target reliability [5]. mMTC needs to support a large number of connected users in a given coverage area with limited spectrum resources. By integrating these two business categories, the concept of massive URLLC (mURLLC) was proposed in [6]. However, its stringent access latency and reliability requirements bring serious challenges to emerging massive access, as it requires simultaneously ensuring latency, reliability, and massive access requirements. mURLLC cannot be achieved solely by introducing new modulation and coding schemes or increasing channel bandwidth and the number of transmit/receive antennas.

Fortunately, more emerging access technologies are being considered as potential solutions to these challenges [7], [8]. Specifically, to further reduce the access latency of mURLLC, grant-free random access (GFRA) was proposed and has been considered as a promising alternative since it eliminates the need to wait for the grant and assigned pilot sequence from the base station (BS) [9]. For improving access reliability, massive multiple input multiple output (MIMO) technology can enhance the access reliability due to its good diversity gain [10]. Cross-layer techniques and retransmission-based technologies can also improve the access reliability [11]. Moreover, the network architecture holds significant potential for improving system reliability and capacity to support mURLLC access. Recent research shows that cell-free massive MIMO (CF-mMIMO) systems can greatly improve the quality of service (QoS) through cooperative processing [12], and are envisioned as alternatives to traditional cellular and multi-cell systems. However, the performance gain of the separate emerging potential access technologies alone cannot satisfy the requirements of mURLLC. Designing a mURLLC access framework that leverages the advantages of integrating existing access technologies is crucial. Therefore, in this context, we

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investigate massive GFRA in CF-mMIMO URLLC systems.

A. Related work

For access reliability, Polyanskiy derived the approximate achievable rate of the system by considering the channel dispersion with given SNR and bit error rate in the finite block-length (FBL) regime [13]. This study indicates the relationship among the achievable rate, transmission block length, bandwidth and decoding error probability, providing a theoretical foundation for subsequent mURLLC access research. In [14], the authors proposed a pioneering space-time two-dimensional channel coding method for reliable transmission with low delay by leveraging the spatial freedom of the massive MIMO system. Additionally, [15] investigated the successful symbol transmission rate for GFRA in massive MIMO systems and improved this performance metric by optimizing the system access parameters and pilot length. Furthermore, [16] investigated GFRA in massive MIMO system and derived the closed-form expressions for the success probability of GFRA with different beamforming technologies.

In addition, some research has improved the reliability of mURLLC access by repetition-based transport protocols and multiple connectivity. Reference [17] proposed a repetition-based non-orthogonal multiple access (NOMA) scheme and derived the upper bound of the outage probability by leveraging the high diversity of repetition transmission. In [18], the performance of the GFRA-based NOMA in URLLC IoT network was investigated, where packet replicas-based diversity transmission was adopted to enhance URLLC requirements. Furthermore, the authors analyzed the reliability performance of the GFRA scheme with a different number of preambles for mMTC scenarios [19], and further extended it to a case that considering repetition and retransmission to achieve performance trade-off among the number of preambles, repetitions, and retransmissions. Reference [20] proposed a mathematical model to maximize the average success probability and optimize the repetition rate in a single cell.

On the other hand, most research on latency has focused on the End-to-End (E2E) latency of users in cellular networks, while few references investigated access latency. First, reference [21] pointed out the components of E2E latency and the trade-off relationship among latency, reliability, and other performance metrics of 6G. To further reduce latency, finite block-length theory was proposed to apply to short packet transmission. According to the numerology specification of 5G new radio, shorter data packet configurations can lead to shorter transmission time intervals (TTI), thereby further reducing transmission latency [13]. Based on this, a physical-layer frame structure compatible with GF-NOMA was proposed in [22], and further investigated success probability, latency and energy efficiency of users with different physical-layer parameters. Moreover, a joint block-length and power allocation method for energy-efficient URLLC in the IoT network was proposed to maximize energy efficiency while meeting URLLC constraints (i.e., reliability and latency) [23]. Except for transmission latency, queuing latency was investigated in [24]. Furthermore, an adaptive block-length scheme

satisfying URLLC requirements for multiple access scenarios was proposed in [25], which aims to achieve a trade-off between transmission latency and queuing latency. Based on this, [26] extended it to cell-free scenarios and investigated the E2E latency of the K -repetitions-based GFRA scheme.

B. Motivation and Main Contributions

On the one hand, although some literature have studied URLLC, most studies only focus on separate performance metrics of URLLC, such as reliability, latency or energy efficiency rather than focusing on mURLLC access. On the other hand, although the success probability of grant-free random access with massive MIMO has investigated in some literature [2], few references integrate URLLC with massive access to investigate access reliability and latency in CF-mMIMO system. Motivated by the above issues, we mainly investigate the access reliability and latency performance for massive access in the CF-mMIMO URLLC system. The following two core questions are tried to answer. 1) *How to model the massive GFRA framework in CF-mMIMO URLLC system, and how can access reliability be evaluated and improved for mURLLC access?* 2) *How can access latency be analyzed and reduced for massive access in CF-mMIMO URLLC systems, while satisfying stringent reliability constraints?* The major contributions of this paper are summarized as follows:

- Distinguished from existing work [2], leveraging macro diversity of the CF-mMIMO system and considering the massive access procedure and associated GFRA protocol, we propose a massive GFRA framework based on the CF-mMIMO URLLC system, consisting of access reliability evaluation and access latency analysis. This access model supports mURLLC access due to the macro diversity and user centric-based scalability of the CF-mMIMO system.
- Based on the proposed CF massive GFRA model and integrating the FBL theory, we analyze the approximated distribution of signal to interference plus noise power (SINR) and derive the approximated closed-form expressions of the decoding error probability and outage probability for random access users in the FBL regime by using the approximated distribution of SINR.
- By comprehensively considering massive GFRA procedure, we develop a comprehensive access reliability evaluation model. First, we derive the expression of access success probability of active users based on pilot collision probability, outage probability, and FBL decoding error probability. We then evaluate access reliability of the attempted users and jointly optimize the pilot length and data block-length by maximizing the access success probability of the active user. Furthermore, leveraging the macro diversity of the CF-mMIMO system, we propose a user centric-based multiple APs scheme to improve access reliability.
- Integrating FBL transmission theory, packet queuing theory and associated GFRA protocol, we analyze the access latency for massive access in the CF-mMIMO URLLC system, which consists of propagation latency, processing latency, queuing latency and transmission latency.

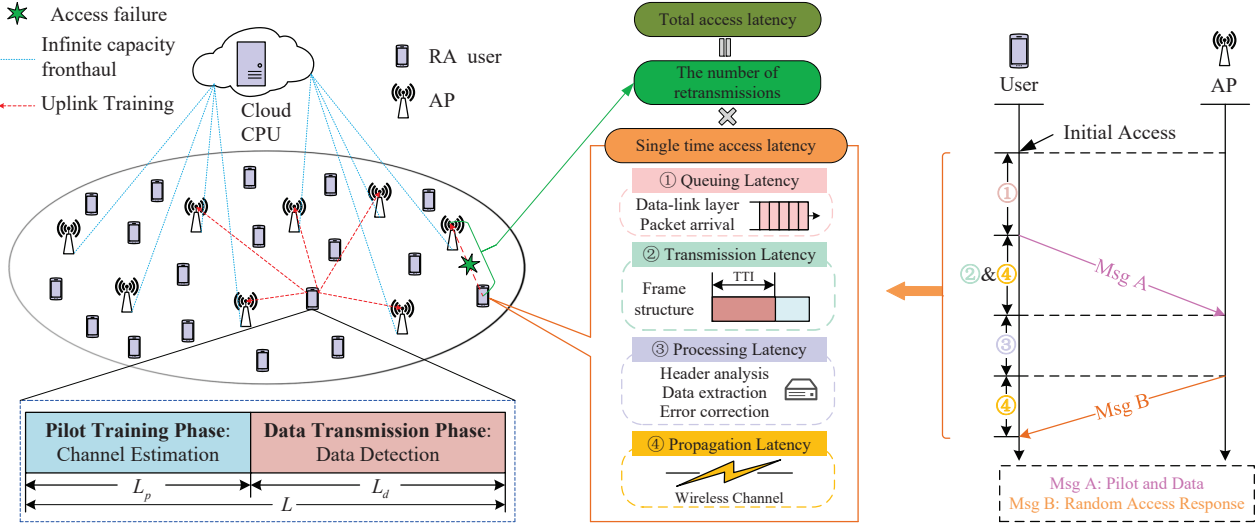


Fig. 1. The illustration of the two-phase grant-free random access procedure in CF-mMIMO system and access latency components of attempted access user.

Furthermore, we develop a retransmission-based access latency model incorporating access failure. By jointly considering random access reliability and the number of retransmissions, we evaluate the access latency performance for massive access.

- Our convincing simulation results demonstrate the rationality and effectiveness of the CF massive access model for analyzing the access reliability and latency. Moreover, simulations verify the trade-off between pilot length and data block-length, and the CF-mMIMO system has better macro diversity and is more scalable to support the massive access.

C. Paper Organization and Notations

The reminder of this paper is organized as follows. Section II introduces the system model. Section III studies performance analysis of the GFRA based on CF mMIMO system, which consists of access reliability, access latency, trade-off between pilot length and block-length. Simulation results are provided in Section IV. Conclusions are drawn in Section V.

Throughout this paper, $(\cdot)^T$ denote transpose $\mathbb{E}(\cdot)$ denotes expectation operation and $\mathbb{V}(\cdot)$ denotes variance. $|\cdot|$ denotes the determinant. $\text{Tr}(\cdot)$ denotes the trace operation. $\Gamma(N, 1)$ denotes the Γ distribution with N freedom, whose PDF is $f(x) = \frac{x^{N-1}}{(N-1)!} e^{-x}$. $\mathcal{N}(\mu, \sigma^2)$ ($\mathcal{CN}(\mu, \sigma^2)$) denotes a (complex) Gaussian distribution with mean μ and variance σ^2 .

II. SYSTEM MODEL

A. System Model

In this paper, we consider an uplink CF-mMIMO system that includes M arbitrarily distributed APs, K users, and a central computing unit (CPU). As Fig. 1, each AP is equipped with N antennas and jointly serves K single-antenna users in the coverage area. All M APs are connected to the CPU via

infinite capacity fronthauls¹, allowing the received signals at the APs to be collected and jointly processed at the CPU. We consider that all APs and antennas are fully separated and independent. Furthermore, the block fading channel model is adopted [27]. The channel gain between the n -th antenna at the m -th AP and the k -th user is given by $g_{mnk} = \sqrt{\beta_{mk}} h_{mnk}$, where β_{mk} denotes the large-scale fading coefficient (LSFC) between the k -th user and the m -th AP, which generally is related to path-loss and shadowing. Specially, in this paper, the LSFCs are assumed to be known at the CPU [28]. And $h_{mnk} \sim \mathcal{CN}(0, 1)$ denotes the small-scale fading coefficient.

Then, the channel vector from the k -th user to all APs can be represented as $\mathbf{h}_k = [\mathbf{h}_{1k}^T, \mathbf{h}_{2k}^T, \dots, \mathbf{h}_{Mk}^T]^T \in \mathbb{C}^{MN \times 1}$, where $\mathbf{h}_{mk} = [g_{m1k}, g_{m2k}, \dots, g_{mNk}]^T \in \mathbb{C}^{N \times 1}$ denotes the channel vector between the m -th AP and the k -th user. In this paper, to reduce access latency, the GFRA protocol is adopted, which is different from the grant-based access protocol, as Fig. 1 (right figure), once a user is activated, it directly transmits Msg A (a preamble and data) without waiting grant and receives Msg B (random access response) from BS. Msg A's frame is illustrated as Fig. 1 (left figure), where L slots are divided into the pilot phase of duration L_p and the data phase of duration L_d . We primarily consider a centralized cell free paradigm to deal with the local received signal [27]. In other words, all K users transmit pilot or data signals to the APs, which then parallel transmit local received signals to the CPU. The CPU subsequently performs channel estimation and data detection. We assume that the transmissions for all devices are well synchronized.

B. Uplink Pilot Training and Channel Estimation

In this paper, we assume that the CPU first pre-assigns mutually orthogonal L_p -length pilot sequence to each active

¹This assumption enables tractable analysis and focuses attention on radio-layer challenges. The finite capacity backhaul is beyond the scope of this paper and can be considered a topic for future research.

user, which can be denoted by $\mathbf{p}_k = [p_{k,1}, p_{k,2}, \dots, p_{k,L_p}]^T$ with $\|\mathbf{p}_k\|^2 = L_p$. Let $L_p \leq \tau_c$, where τ_c denotes the number of the channel use in coherence time. The uplink received pilot signal $\mathbf{y}_m^p \in \mathbb{C}^{N \times L_p}$ at the m -th AP can be expressed as

$$\mathbf{y}_m^p = \sum_{k=1}^K \sqrt{\rho_k} \mathbf{h}_{mk} \mathbf{p}_k^T + \mathbf{w}_m \in \mathbb{C}^{N \times L_p}, \quad (1)$$

where ρ_k denotes the transmit power of the k -th user, $\mathbf{w}_m \sim \mathcal{CN}(0, \sigma^2)$ is normalized additive white Gaussian noise.

Then, all the received signals at each AP are transmitted to the CPU to perform local channel estimation for each user. First, the correlation operation is performed by itself normalized pilot signal $\mathbf{p}_k / \sqrt{L_p}$, which can be expressed as

$$\begin{aligned} \tilde{\mathbf{y}}_m^p &= \mathbf{y}_m^p \frac{\mathbf{p}_k^*}{\sqrt{L_p}} = \sum_{k=1}^K \frac{\sqrt{\rho_k}}{\sqrt{L_p}} \mathbf{h}_{mk} \mathbf{p}_k^T \mathbf{p}_k^* + \mathbf{w}_m \frac{\mathbf{p}_k^*}{\sqrt{L_p}} \\ &= \sum_{k \in \mathcal{U}_k} \sqrt{\rho_k L_p} \mathbf{h}_{mk} + \tilde{\mathbf{w}}_m \in \mathbb{C}^N, \end{aligned} \quad (2)$$

where $\mathcal{U}_k$ denotes the index set of users selected the same pilot sequence. And $\tilde{\mathbf{w}}_m = \frac{1}{\sqrt{L_p}} \mathbf{w}_m \mathbf{p}_k^* \sim \mathcal{CN}(0, \sigma^2)$, since the pilot sequence is deterministic and $\|\mathbf{p}_k\|^2 = L_p$.

Remark 1: $|\mathcal{U}_k|$ will affect the channel estimation. If $|\mathcal{U}_k| > 1$, $\hat{\mathbf{h}}_{mk}$ will contain the channel interference from non-desired user with same pilot sequence, namely, pilot contamination. If $|\mathcal{U}_k| = 1$, the channel interference will vanish. However, in general, the number of users is greater than the number of pilot sequences. So, the access reliability under the channel estimation error and pilot sequence collision will be discussed in the follow section.

Based on the minimum mean square error (MMSE) channel estimation criterion [27], the MMSE channel estimation² between the k -th user to the m -th AP can be expressed as

$$\hat{\mathbf{h}}_{mk} = \sqrt{\rho_k L_p} \mathbf{R}_{mk} \Phi_{mk}^{-1} \tilde{\mathbf{y}}_m^p. \quad (3)$$

where, $\mathbf{R}_{mk} = \beta_{mk} \mathbf{I}_N$ denotes the spatial correlation matrix. $\Phi_{mk} = \mathbb{E}\{\tilde{\mathbf{y}}_m^p (\tilde{\mathbf{y}}_m^p)^H\} = \sum_{k \in \mathcal{U}_k} \rho_k L_p \mathbf{R}_{mk} + \sigma^2 \mathbf{I}_N$ denotes the correlation matrix of the received signal.

Then, we analysis the statistical characteristics of $\hat{\mathbf{h}}_{mk}$, which can be represented as

$$\begin{aligned} \mathbf{C}_{\hat{\mathbf{h}}_{mk}} &= \mathbb{E}\{\rho_k L_p \mathbf{R}_{mk} \Phi_{mk}^{-1} \tilde{\mathbf{y}}_m^p (\tilde{\mathbf{y}}_m^p)^H \Phi_{mk}^{-1} \mathbf{R}_{mk}\} \\ &= \rho_k L_p \mathbf{R}_{mk} \Phi_{mk}^{-1} \mathbb{E}\{\tilde{\mathbf{y}}_m^p (\tilde{\mathbf{y}}_m^p)^H\} \Phi_{mk}^{-1} \mathbf{R}_{mk} \\ &= \rho_k L_p \mathbf{R}_{mk} \Phi_{mk}^{-1} \mathbf{R}_{mk} \\ &= \frac{\rho_k L_p \beta_{mk}^2}{\sum_{k \in \mathcal{U}_k} \rho_k L_p \beta_{mk} + \sigma^2} \mathbf{I}_N. \end{aligned} \quad (4)$$

Assume that the channel estimation error is defined as $\tilde{\mathbf{h}}_{mk} = \mathbf{h}_{mk} - \hat{\mathbf{h}}_{mk}$, and the estimated channel $\hat{\mathbf{h}}_{mk}$ and channel estimation error $\tilde{\mathbf{h}}_{mk}$ are independent respectively. Based on the statistical characteristics of $\hat{\mathbf{h}}_{mk}$, we can know

²The MMSE channel estimation require to calculate the correlation matrix, but which can provide a better channel estimation performance. The least squares (LS) estimator is also an alternative method, and the LS don't require the prior statistical characteristics.

that the distribution of channel estimation error $\tilde{\mathbf{h}}_{mk}$ yields $\tilde{\mathbf{h}}_{mk} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}_{\tilde{\mathbf{h}}_{mk}})$, concretely,

$$\begin{aligned} \mathbf{C}_{\tilde{\mathbf{h}}_{mk}} &= \mathbb{E}\{\tilde{\mathbf{h}}_{mk} \tilde{\mathbf{h}}_{mk}^H\} = \mathbb{E}\{(\mathbf{h}_{mk} - \hat{\mathbf{h}}_{mk})(\mathbf{h}_{mk} - \hat{\mathbf{h}}_{mk})^H\} \\ &= \mathbb{E}\{\mathbf{h}_{mk} \mathbf{h}_{mk}^H\} - \mathbb{E}\{\hat{\mathbf{h}}_{mk} \hat{\mathbf{h}}_{mk}^H\} = \mathbf{R}_{mk} - \mathbf{C}_{\hat{\mathbf{h}}_{mk}} \\ &= \frac{\sum_{k \in \mathcal{U}_k \setminus k} \rho_k L_p \beta_{mk}^2 + \sigma^2 \beta_{mk}}{\sum_{k \in \mathcal{U}_k} \rho_k L_p \beta_{mk} + \sigma^2} \mathbf{I}_N \\ &\stackrel{(a)}{=} \frac{\sigma^2 \beta_{mk}}{\rho_k L_p \beta_{mk} + \sigma^2} \mathbf{I}_N \end{aligned} \quad (5)$$

where $\{\mathcal{U}_k \setminus k\}$ denotes the index k is eliminated in the set $\mathcal{U}_k$. Step (a) holds because the pilot contamination is not considered.

Remark 2: First, we only consider the case where the user has not experienced pilot contamination, namely $|\mathcal{U}_k| = 1$, then the channel estimation error matrix is simplified as $\frac{\sigma^2 \beta_{mk}}{\rho_k L_p \beta_{mk} + \sigma^2} \mathbf{I}_N$. Second, the channel estimation error matrix $\mathbf{C}_{\tilde{\mathbf{h}}_{mk}}$ is only related to the LSFCs. It's worth noting that LSFCs are significantly different between user and different APs. So, for the user k , the contribution of each AP to $\mathbf{C}_{\tilde{\mathbf{h}}_{mk}}$ varies, which in turn affects the access success probability.

C. Uplink Data Transmission

Recall that the two-phase GFRA protocol is applied in this paper. In the data transmission phase, the data from the user is transmitted simultaneously to all APs. Therefore, the received signal at the m -th AP can be expressed as

$$\mathbf{y}_m^d = \sqrt{\rho_d} \sum_{k=1}^K \mathbf{h}_{mk} \mathbf{s}_k + \mathbf{z}_m, \quad (6)$$

where ρ_d denotes the normalized SNR, $\mathbf{s}_k$ denotes the transmitted data of the k -th user, i.e., QPSK modulation symbol, which satisfies $\|\mathbf{s}_k\|^2 = 1$. And $\mathbf{z}_m \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$ denotes the normalized background noise at the m -th AP. Due to the considered centralized cell free paradigm, the collection received signal at all AP can be denoted as

$$\mathbf{Y}^d = \sum_{m=1}^M \sum_{k=1}^K \sqrt{\rho_d} \mathbf{h}_{mk} \mathbf{s}_k + \mathbf{Z} = \sum_{k=1}^K \sqrt{\rho_d} \mathbf{h}_k \mathbf{s}_k + \mathbf{Z}, \quad (7)$$

where $\mathbf{Z} = [\mathbf{z}_1^T, \mathbf{z}_2^T, \dots, \mathbf{z}_M^T]^T \in \mathbb{C}^{MN \times 1}$. Based on Eq. (7), by applying the combining vector $\mathbf{r}^3$ for all the APs, we can

³The received combining vector can be arbitrary, such as maximum ratio combining (MRC), zero-forcing (ZF) and MMSE. Since MMSE combining includes additional noise regularization term and balances noise enhancement and interference suppression by minimizing the mean squared error between the transmitted and received signals, it achieves optimal performance but with the highest complexity $\mathcal{O}\left(\frac{3K^2 MN}{2} + \frac{KMN}{2} + \frac{K^3 - K}{3}\right)$. MRC has the poorest performance but is a scalable scheme with the lowest complexity $\mathcal{O}(MN)$. ZF neglects the noise regularization term and performs multi-user interference cancellation by using pseudo-inverse operation. It can achieve a sub-optimal performance with complexity similar to MMSE, $\mathcal{O}\left(\frac{3K^2 MN}{2} + \frac{KMN}{2} + \frac{K^3 - K}{3}\right)$. Thus, MRC can achieve a better performance-complexity trade-off. In this work, considering the performance-complexity trade-off, scalability, applicability to massive URLLC grant-free scenarios and analytical tractability, we adopt MRC as the combining vector.

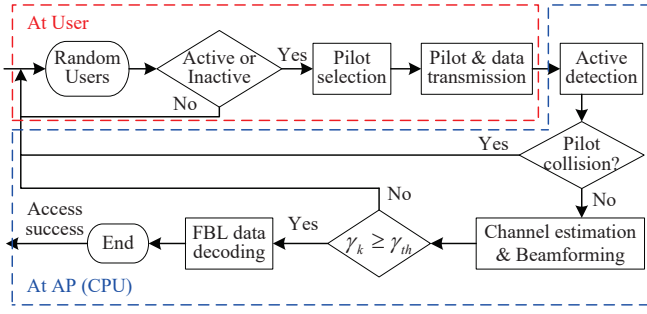


Fig. 2. Procedure of grant-free random access for CF-mMIMO systems.

know the received signal of the k -th user can be represented as

$$\begin{aligned}
 \mathbf{y}_k^d &= \mathbf{r}_k^H \mathbf{Y}^d = \mathbf{r}_k^H \sum_{m=1}^M \sum_{k=1}^K \sqrt{\rho_d} \mathbf{h}_{mk} \mathbf{s}_k + \mathbf{Z} = \underbrace{\sqrt{\rho_d} \sum_{m=1}^M \mathbf{r}_k^H \hat{\mathbf{h}}_{mk} \mathbf{s}_k}_{\text{desired signal}} \\
 &+ \underbrace{\sum_{i \neq k} \sum_{m=1}^M \mathbf{r}_k^H \mathbf{h}_{mi} \mathbf{s}_i}_{\text{other user interference}} + \underbrace{\sum_{m=1}^M \sqrt{\rho_d} \mathbf{r}_k^H \tilde{\mathbf{h}}_{mk} \mathbf{s}_k}_{\text{estimation error}} + \underbrace{\sum_{m=1}^M \mathbf{r}_k^H \mathbf{Z}}_{\text{noise interference}} \\
 &= \underbrace{\sqrt{\rho_d} \mathbf{r}_k^H \hat{\mathbf{h}}_k \mathbf{s}_k}_{\text{desired signal}} + \underbrace{\sum_{i \neq k} \mathbf{r}_k^H \mathbf{h}_i \mathbf{s}_i}_{\text{other user interference}} + \underbrace{\sqrt{\rho_d} \mathbf{r}_k^H \tilde{\mathbf{h}}_k \mathbf{s}_k}_{\text{estimation error}} + \underbrace{\mathbf{r}_k^H \mathbf{Z}}_{\text{noise}}
 \end{aligned} \tag{8}$$

where, $\mathbf{h}_i$ denotes the channel collection vector from the k -th user to all APs. By exploiting the last equation, we can derive the SINR of the k -th user [29], which can be expressed as

$$\begin{aligned}
 \gamma_k &= \frac{\rho_d |\mathbf{r}_k^H \hat{\mathbf{h}}_k \mathbf{s}_k|^2}{\rho_d \left| \sum_{i \neq k} \mathbf{r}_k^H \mathbf{h}_i \mathbf{s}_i \right|^2 + \rho_d |\mathbf{r}_k^H \tilde{\mathbf{h}}_k \mathbf{s}_k|^2 + |\mathbf{r}_k^H \mathbf{Z}|^2} \\
 &= \frac{\rho_d |\mathbf{r}_k^H \hat{\mathbf{h}}_k \mathbf{s}_k|^2}{\rho_d \left| \sum_{i \neq k} \mathbf{r}_k^H \mathbf{h}_i \mathbf{s}_i \right|^2 + \rho_d |\mathbf{r}_k^H \tilde{\mathbf{h}}_k \mathbf{s}_k|^2 + |\mathbf{r}_k^H \mathbf{Z}|^2} \\
 &\stackrel{(a)}{=} \frac{\rho_d |\mathbf{r}_k^H \hat{\mathbf{h}}_k \mathbf{s}_k|^2}{\rho_d \left| \sum_{i \neq k} \mathbf{r}_k^H \mathbf{h}_i \mathbf{s}_i \right|^2 + \rho_d \left| \sum_{i=1}^K \mathbf{C}_{\tilde{\mathbf{h}}_{mi}} + \sigma^2 \mathbf{I}_{MN} \right| \mathbf{r}_k} \\
 &\stackrel{(b)}{=} \frac{|\hat{\mathbf{h}}_k|^2}{\sum_{i \neq k} \frac{|\mathbf{h}_i^H \mathbf{h}_i|^2}{|\hat{\mathbf{h}}_k|^2} + \text{Tr}(\mathbf{C}_{\tilde{\mathbf{h}}_k}) + \frac{\sigma^2}{\rho_d}}
 \end{aligned} \tag{9}$$

where (a) holds due to $\mathbf{C}_{\tilde{\mathbf{h}}_k} = \sum_{i=1}^K \mathbf{C}_{\tilde{\mathbf{h}}_{mi}}$, the step (b) holds due to $\mathbf{r}_k = \hat{\mathbf{h}}_k$. It's worth noting that we assume that once the user's pilot collides, the attempted access user is considered to have failed. Therefore, the derived effective γ_k only contains the case where the pilot was not contaminated, which is derived from simplification of Eq. (5) and involves the channel estimation error that no pilot contamination.

III. PERFORMANCE ANALYSIS GRANT FREE RANDOM ACCESS BASED ON CF mMIMO SYSTEM

A. Reliability Analysis of User Access Network

In order to measure the access reliability of the user, we analyze the access success probability. As shown in Fig. 2, according to the GFRA procedure, the access success probability of a user is determined by collision, channel conditions (outage probability) and decoding error probability [16]. We think that the k -th user satisfied with three criteria can successfully access the network. 1) There is no pilot collision⁴ for the k -th user. 2) For the k -th user, $k \in [1, K]$, $\gamma_k \geq \gamma_{th}$, where γ_{th} denotes the SINR threshold for the k -th user. 3) No decoding error occurs. In other words, the access success probability of the k -th user depends on whether the pilot sequence collides, SINR and decoding error probability. Therefore, the access success probability of the k -th user can be represented as

$$\mathbb{P}^{(k)} = \mathbb{P}_{NC}^k \mathbb{P}(\gamma_k \geq \gamma_{th}) (1 - \varepsilon_k), \tag{10}$$

where $\mathbb{P}_{NC}^k$ denotes the pilot non-collision probability for the k -th user. $\mathbb{P}(\gamma_k \geq \gamma_{th})$ represents the probability of the event that γ_k exceeds the SINR threshold, ε_k is the decoding error probability of the k -th user. It is worth noting that the access reliability model only takes into account that access is considered to fail once a pilot collision occurs, and other collision resolutions are not within the scope of consideration.

1) *Pilot non-collision probability of the k -th user:* In considered mURLLC access scenarios, the number of pilots is less than the total number of users, i.e., $(L_p < K)$. A pilot collision occurs when more than one user selects the same pilot. Each pilot is selected independently among all users, which means that the selection process for each pilot is approximately as K independent repeated trials with a probability $1/L_p$. At present, we assume that the number of users selecting the j -th pilot is $\mathcal{N}_j$. If the j -th pilot is only selected by the k -th user, then the k -th user is considered to have no pilot collision. This event is represented by $\Pr\{\mathcal{N}_j = 1\}$, then the probability $\mathbb{P}_j^{(1)}$ of $\Pr\{\mathcal{N}_j = 1\}$ can be expressed as

$$\mathbb{P}_j^{(1)} = \frac{1}{L_p} \left(1 - \frac{1}{L_p}\right)^{K-1}. \tag{11}$$

Therefore, for all available pilots, the probability that the k -th user selects any pilot that does not collide with other users can be considered as the probability of the sum event of $\Pr\{\mathcal{N}_j = 1\}, j \in [1, L_p]$, which can be expressed as

$$\begin{aligned}
 \mathbb{P}_{NC}^k &= \Pr\{\mathcal{N}_1=1\} \cup \Pr\{\mathcal{N}_2=1\} \cup \dots \cup \Pr\{\mathcal{N}_{L_p}=1\} \\
 &= \sum_{j=1}^{L_p} \mathbb{P}_j^{(1)} = \sum_{j=1}^{L_p} \frac{1}{L_p} \left(1 - \frac{1}{L_p}\right)^{K-1} \\
 &= \left(1 - \frac{1}{L_p}\right)^{K-1}.
 \end{aligned} \tag{12}$$

⁴When multiple random access users select the same preamble simultaneously, the BS is unable to acquire the correct channel responses of these random access users and their data transmissions would be failed.

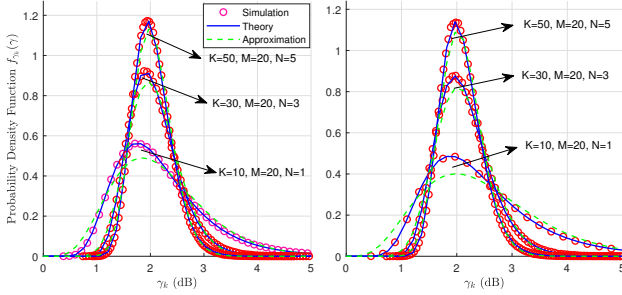


Fig. 3. PDF of SINR in the CF-mMIMO system with different K and number of antennas N for SNR = 0, 10 dB, respectively.

2) *The derivation of $\mathbb{P}(\gamma_k \geq \gamma_{th})$:* From (9), we can know that the simplified expression for γ_k is intractable due to involving correlated Gamma random variable in the numerator and denominator of γ_k . Therefore, it is challenging to obtain the cumulative distribution function (CDF) of γ_k . Most existing literature addresses this challenge by adopting Gamma and Log-normal random variable with the moment matching technique [30]. In this paper, we discovered the numerator and denominator of γ_k yields logarithmic normal distribution by extensive simulations. In other words, the logarithmic operation of the numerator and denominator of γ_k yields normal distribution, as illustrated in Fig. 3. Thus, we can further derive $\mathbb{P}(\gamma_k \geq \gamma_{th})$ by leveraging this approximated Log-normal distribution of the numerator and denominator of γ_k .

In this paper, we adopt the MRC, i.e., $\mathbf{r}_k = \hat{\mathbf{h}}_k$. Let $\Psi_k = \rho_d \left| \sum_{i \neq k}^K \hat{\mathbf{h}}_i^H \hat{\mathbf{h}}_i \right|^2 + \hat{\mathbf{h}}_k^H (\rho_d \sum_{i=1}^K \mathbf{C}_{\hat{\mathbf{h}}_{m,i}} + \sigma^2 \mathbf{I}_{MN}) \hat{\mathbf{h}}_k$, $\Phi_k = \rho_d \left(\left| \hat{\mathbf{h}}_k \right|^2 \right)^2$. So $\gamma_k = \frac{\rho_d \left| \sum_{i \neq k}^K \hat{\mathbf{h}}_i^H \hat{\mathbf{h}}_i \right|^2 + \hat{\mathbf{h}}_k^H (\rho_d \sum_{i=1}^K \mathbf{C}_{\hat{\mathbf{h}}_{m,i}} + \sigma^2 \mathbf{I}_{MN}) \hat{\mathbf{h}}_k}{\rho_d \left(\left| \hat{\mathbf{h}}_k \right|^2 \right)^2}$. Then, we approximate Φ_k and Ψ_k as Log-normal distribution by using moment matching technique, respectively. Namely, $\Phi_k \sim \mathcal{LN}(\mu_{\Phi_k}, \sigma_{\Phi_k}^2)$ and $\Psi_k \sim \mathcal{LN}(\mu_{\Psi_k}, \sigma_{\Psi_k}^2)$. So we can obtain follow **Theorem 1**.

Theorem 1: The logarithmic operation of γ_k yields normal distribution, which can be expressed as

$$\log(\gamma_k) \sim \mathcal{N}(\mu_{\gamma_k}, \sigma_{\gamma_k}^2), \quad (13)$$

where μ_{γ_k} and σ_{γ_k} denotes mean and variance, respectively.

$$\mu_{\gamma_k} = \mu_{\Phi_k} - \mu_{\Psi_k} \text{ and } \sigma_{\gamma_k}^2 = \sigma_{\Phi_k}^2 + \sigma_{\Psi_k}^2 - 2 \log \left(\frac{\mathbb{E}[\Phi_k \Psi_k]}{\mathbb{E}[\Phi_k] \mathbb{E}[\Psi_k]} \right).$$

Proof: Please see Appendix A $\square$

Based on **Theorem 1**, we can further derive $\mathbb{P}(\gamma_k \geq \gamma_{th})$, which can summarized as **Theorem 2**.

Theorem 2: For given a SINR threshold γ_{th} , the probability of the event that γ_k is greater than γ_{th} can be represented as

$$\mathbb{P}(\gamma_k \geq \gamma_{th}) \approx 1 - \frac{1}{2} \text{erfc} \left(- \frac{\ln(\gamma_{th}) - \mu_{\gamma_k}}{\sqrt{2} \sigma_{\gamma_k}} \right). \quad (14)$$

where $\text{erfc}(\cdot)$ is the complementary error function.

Proof: Please see Appendix B. $\square$

3) *The decoding error probability for GFRA in FBL regime:* According to Shannon information theory, the achievable rate of user k can be expressed as $R = \ln(1 + \gamma_k)$. However, it is no longer suitable for URLLC systems where the small data

packets are transmitted. To conquer this problem, some literature has proposed an alternative scheme that approximates the user's achievable rate by the finite block-length theory [13], for a given finite block-length L_d , which can be represented as

$$\tilde{R} = \log_2(1 + \gamma_k) - \sqrt{\frac{V(\gamma_k)}{L_d}} Q^{-1}(\varepsilon), \quad (15)$$

where $Q^{-1}(\cdot)$ denotes the inverse of the Q function, $Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} \exp(-\frac{t^2}{2}) dt$. And $V(\gamma_k) = (1 - \frac{1}{(1+\gamma_k)^2}) (\log_2 e)^2$ denotes channel dispersion. ε is decoding error probability.

From Fig. 1, in the uplink data transmission phase, b bits data is transmitted by L_d channels. So the uplink available rate is $R = b/L_d$. The decoding error probability can be further expressed as

$$\begin{aligned} \varepsilon_k &= \mathbb{E}_{\gamma_k} \left\{ Q \left(\sqrt{\frac{L_d}{V(\gamma_k)}} \left[\log_2(1 + \gamma_k) - \frac{b}{L_d} \right] \right) \right\} \\ &= \int_0^\infty Q \left(\sqrt{\frac{L_d}{V(s)}} \left[\log_2(1 + s) - \frac{b}{L_d} \right] \right) f_{\gamma_k}(s) ds, \end{aligned} \quad (16)$$

where $f_{\gamma_k}(s)$ denotes the probability density function (PDF) of γ_k . It's worth noting that Q function is intractable. In order to better tackle (16), we further approximate Q function as $Q \left(\sqrt{\frac{L_d}{V(\gamma_k)}} \left[\log_2(1 + \gamma_k) - \frac{b}{L_d} \right] \right) \approx \varphi(\gamma_k)$, where $\varphi(\gamma_k)$ can be expressed as [11]

$$\varphi(\gamma_k) = \begin{cases} 1, & \gamma_k \leq \phi_1 \\ \frac{1}{2} - \psi \sqrt{L_d} (\gamma_k - \theta), & \phi_1 < \gamma_k < \phi_2 \\ 0, & \gamma_k \geq \phi_2 \end{cases} \quad (17)$$

where $\phi_1 = \theta - \frac{\sqrt{L_d}}{2\psi}$, $\phi_2 = \theta + \frac{\sqrt{L_d}}{2\psi}$, $\theta = 2^{\frac{b}{L_d}} - 1$ and $\psi = \frac{1}{2\pi \sqrt{2^{\frac{b}{L_d}} - 1}}$. By the above mentioned approximation, Eq. (16) can be further represented as

$$\varepsilon_k \approx \mathbb{E}_{\gamma_k} \{ \varphi(\gamma_k) \} = \psi \sqrt{L_d} \int_{\phi_1}^{\phi_2} F_{\gamma_k}(s) ds, \quad (18)$$

where $F_{\gamma_k}(s) = \mathbb{P}\{\gamma_k \leq s\}$ denotes the CDF of γ_k . The proof of Eq. (18) can be found in Appendix C.

Obviously, Eq. (18) is still not a closed-form solution, which is intractable to observe the trade-off between pilot length and data block length in the FBL regime. Therefore, we further derive the closed-form expression in theorem 3.

Theorem 3: Based on the theorem 1 and Eq. (18), the closed-form solution of ε_k can be expressed as

$$\begin{aligned} \varepsilon_k &\approx \mathbb{E} \{ \varphi(\gamma_k) \} = \psi \sqrt{L_d} \int_{\phi_1}^{\phi_2} F_{\gamma_k}(s) ds \\ &= \frac{\psi \sqrt{L_d}}{D} \left[(u_2 - u_1) - \sum_{n=0}^{MN-1} (MN-n) \tilde{\Upsilon}_n \right]. \end{aligned} \quad (19)$$

where $\tilde{\Upsilon}_n = \frac{u_1^n}{n!} e^{-u_1} - \frac{u_2^n}{n!} e^{-u_2}$ and $u_1 = D\phi_1$, $u_2 = D\phi_2$. $D = \text{Tr}(\mathbf{C}_{\hat{\mathbf{h}}_k}) + \frac{\sigma^2}{\rho_d}$.

Proof: Please see Appendix D. $\square$

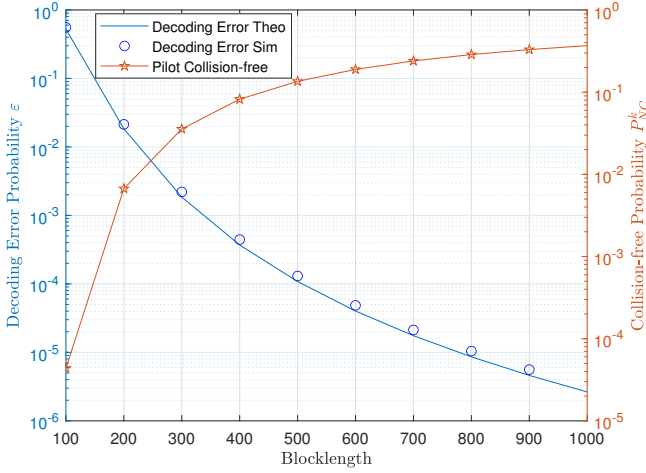


Fig. 4. Monotonicity illustration of decoding error probability and pilot collision-free probability ($\mathbb{P}_{NC}^k$) vs. block-length with 32-byte transmission.

Therefore, the access success probability of the k -th user can be rewritten as

$$\begin{aligned} \mathbb{P}^{(k)} &= \mathbb{P}_{NC}^k * \mathbb{P}(\gamma_k \geq \gamma_{th}) (1 - \varepsilon_k) \\ &= \left(1 - \frac{1}{L_p}\right)^{K-1} \left(1 - \frac{1}{2} \operatorname{erfc}\left(-\frac{\ln(\gamma_{th}) - \mu_{\gamma_k}}{\sqrt{2}\sigma_{\gamma_k}}\right)\right) \\ &\quad \times \frac{\psi\sqrt{L_d}}{D} \left[(u_2 - u_1) - \sum_{n=0}^{MN-1} (MN-n) \tilde{\Upsilon}_n \right]. \end{aligned} \quad (20)$$

B. Joint Optimization of Block-length and Pilot Length

There is a trade-off between pilot length and block-length based on the observation. We then formulate a joint optimization problem that joint optimizes block-length and pilot length with given constraints to maximize the access reliability.

Observation 1: As Fig. 4, the decoding error probability gradually decreases and non-outage probability increases as blocklength increases. For given blocklength, employing a longer training sequence can enhance the accuracy of channel estimation, thereby enabling transmissions with greater reliability. However, this comes at the expense of fewer symbols available for actual data transmission. This dilemma becomes especially noteworthy in the FBL regime. Therefore, there is a trade-off between pilot length and data block length.

Recall that the access success probability of the k -th user is expressed as $\mathbb{P}^{(k)}$. Due to the adopted GFRA, the transmission latency of pilot and data phase is considered. So the transmission latency is constrained. Given frequency resource and latency constraints, the data block-length and pilot length can be jointly optimized by following optimization problem.

$$\max_{L_p, L_d} \mathbb{P}^{(k)} \quad (21)$$

$$\text{subject to } T_t \leq T_{th} \quad (21a)$$

$$\varepsilon \leq \varepsilon_{th} \quad (21b)$$

$$L_p \leq L \quad (21c)$$

$$L_d \leq L \quad (21d)$$

$$L_p, L_d \in \mathbb{N} \quad (21e)$$

Algorithm 1 Joint optimization of pilot length and block-length.

Input: $K, L, T_{th}, \varepsilon_{th}$

- 1: **Initialize:** $L_p^i, L_d^i, \varepsilon_{th}, L_{d,max}, L_{d,min}$, maximum iteration index i_{max} and tolerance δ .
- 2: **while** $\mathbb{P}_i^{(k)} - \mathbb{P}_{i-1}^{(k)} \geq \delta$ **do**
- 3: Update $L_{d,mid}^i = L_{d,max} - L_{d,min}/2, L_p^i = L - L_{d,mid}^i$.
- 4: **if** $\mathbb{P}_i^{(k)}(L_{d,mid}^i, L_p^i, K) \geq \mathbb{P}_i^{(k)}(L_{d,min}^i, L_p^i, K)$ **then**
- 5: Update $L_{d,min}^i = L_{d,mid}^i$
- 6: **else** $L_{d,max}^i = L_{d,mid}^i$
- 7: **end if**
- 8: $L_{d,opt}^i = L_{d,mid}^i$
- 9: Obtain $L_{p,opt}^i$ by $L_p^i = L - L_{d,opt}^i$.
- 10: Update $\mathbb{P}_i^{(k)}(L_{d,opt}^i, L_{p,opt}^i, K)$ by (20)
- 11: $i = i + 1$
- 12: **end while**

Output: $L_{p,opt}, L_{d,opt}$ and $\mathbb{P}_i^{(k)}$.

where T_{th} and ε_{th} denote latency constraint threshold and reliability requirement, respectively. (21a) indicates that the transmission latency constraint is satisfied. (21b) indicates that the reliability requirement constraint is satisfied. (21c) and (21d) hold since the GFRA protocol is adopted, the data block-length L_d and the pilot length L_p are less than the overall transmission block length. Moreover, the data block-length L_d and the pilot length L_p are positive integers. This optimization problem can be solved by Algorithm 1.

It is worth noting that we aim to jointly optimize the pilot length and data block-length by maximizing the access success probability ($\mathbb{P}^{(k)}$). Since the access success probability involves multiple variables, the optimization problem is a typical nonlinear integer programming problem. Due to $L = L_p + L_d$, when given parameters L, K, T_{th} and ε_{th} , the optimization problem has only one independent variable. Therefore, it can be solved by an iterative method in the feasible domain. Furthermore, since we consider URLLC in finite block-length regime and T_{th} is very small, it implies the total block-length L is very small. Meanwhile, due to $L_p, L_d \in \mathbb{N}$, we can conclude that the feasible domain space of the optimization problem (21) is quite limited, and the optimization problem can be solved by an iterative bisection search (Algorithm 1). Therefore, Algorithm 1 can converge to the global optimum, obtaining the optimal pilot length and data block-length given the total block-length L .

C. Access Reliability of User-centric-based Multiple APs

In a CF system, each user can be served by multiple APs around them. The links between each user and corresponding multiple APs experience different fading, which is contributed to enhance access reliability. Motivated by this, considering the macro diversity of the user centric-based CF-mMIMO system, we propose a novel distributed multiple APs scheme to improve the access success probability. The so-called distributed multi-APs scheme is that the received signals at multiple APs are jointly processed in a distributed way, thereby improving the user's access probability by leveraging the diversity gain

of multiple APs. Based on Section II.A, we can further derive the access success probability of any attempted access active user at the m -th AP. First, assume that the received signal is processed by only one AP, the SINR of the k -th user at the m -th AP can be expressed as

$$\gamma_{m,k} = \frac{\rho_d |\hat{\mathbf{h}}_{m,k}|^4}{\rho_d \left| \sum_{i \neq k}^K \hat{\mathbf{h}}_{m,k}^H \hat{\mathbf{h}}_{m,i} \right|^2 + \hat{\mathbf{h}}_{m,k}^H \left(\rho_d \sum_{i=1}^K \mathbf{C}_{\hat{\mathbf{h}}_{m,i}} + \sigma^2 \mathbf{I}_{MN} \right) \hat{\mathbf{h}}_{m,k}}.$$

Then, by leveraging $\gamma_{m,k}$ and **Theorem 2**, we can obtain the non-outage probability of the k -th user at the m -th AP, which is expressed as

$$\bar{P}_{out}^{m,k} \approx 1 - \frac{1}{2} \operatorname{erfc} \left(-\frac{\ln(\gamma_{th}) - \mu_{\gamma_{m,k}}}{\sqrt{2}\sigma_{\gamma_{m,k}}} \right). \quad (22)$$

where $\mu_{\gamma_{m,k}}$ and $\sigma_{\gamma_{m,k}}$ respectively denote the mean and variance of $\gamma_{m,k}$, which has been defined in Eq. (13).

Furthermore, according to Eq. (16) and **Theorem 3**, the decoding error probability $\varepsilon_{m,k}$ of the k -th user at the m -th AP can be represented as

$$\begin{aligned} \varepsilon_{m,k} &\approx \mathbb{E} \{ \varphi(\gamma_{m,k}) \} = \psi \sqrt{L_d} \int_{\phi_1}^{\phi_2} F_{\gamma_{m,k}}(s) ds \\ &= \psi \sqrt{L_d} \left[(\phi_2 - \phi_1) - \frac{1}{D} \sum_{n=0}^{N-1} (N-n) \tilde{\Upsilon}_n \right]. \end{aligned} \quad (23)$$

where $\tilde{\Upsilon}_n = \frac{u_1^n}{n!} e^{-u_1} - \frac{u_2^n}{n!} e^{-u_2}$ and $D = \operatorname{Tr}(\mathbf{C}_{\hat{\mathbf{h}}_k}) + \frac{\sigma^2}{\rho_d}$.

As showed in Fig. 1, since the access success probability of the k -th user is determined by pilot collision probability, channel condition (i.e., $\gamma_{m,k} \geq \gamma_{th}$) and decoding error probability. Taking these factors into account, we derive the access success probability $P_{m,k}$ of the k -th user at the single m -th AP, which is denoted as

$$\begin{aligned} P_{m,k} &= P_{NC}^k \times \bar{P}_{out}^{m,k} \times (1 - \varepsilon_{m,k}) \\ &= \left\{ \left(1 - \frac{1}{L_p} \right)^{K-1} \right\} \left(1 - \frac{1}{2} \operatorname{erfc} \left(-\frac{\ln(\gamma_{th}) - \mu_{\gamma_{m,k}}}{\sqrt{2}\sigma_{\gamma_{m,k}}} \right) \right) \\ &\times \left\{ \psi \sqrt{L_d} \left[(\phi_2 - \phi_1) - \frac{1}{D} \sum_{n=0}^{N-1} (N-n) \tilde{\Upsilon}_n \right] \right\}. \end{aligned} \quad (24)$$

Leveraging the macro diversity of the user centric-based CF-mMIMO system to enhance access reliability. Assume that there are α APs serving the k -th user⁵, meaning that α APs are available for the k -th user to access the network in a distributed manner. Let $\mathcal{A}_i$ denote the event that the k -th user successfully accesses the network through the i -th AP. Thus, the access success probability of the k -th user under α APs can be expressed as follow

$$\begin{aligned} P_k^{(\alpha)} &= \Pr \{ \mathcal{A}_1 \cup \dots \cup \mathcal{A}_\alpha \} = 1 - \Pr \{ \bar{\mathcal{A}}_1 \cap \dots \cap \bar{\mathcal{A}}_\alpha \} \\ &\stackrel{(a)}{=} 1 - \Pr \{ \bar{\mathcal{A}}_1 \} \cap \dots \cap \Pr \{ \bar{\mathcal{A}}_\alpha \} \\ &= 1 - (1 - P_{1,k})(1 - P_{2,k}) \dots (1 - P_{\alpha,k}) \\ &= 1 - \prod_{i=1}^{\alpha} (1 - P_{i,k}), \end{aligned} \quad (25)$$

where $\bar{\mathcal{A}}_i$, $i \in [1, \dots, \alpha]$ denote the opposite event of $\mathcal{A}_i$. The step (a) holds since all α APs are independent of each other and all corresponding events are mutually independent.

⁵In general, α is limited and usually determined by the number of the closest multiple APs, such as reference [28], $\alpha = 3$.

D. Access Latency Performance Analysis

In this section, we aim to investigate the access latency for mURLLC scenarios. According to the specification of 3GPP [31], the user plane E2E latency generally includes wireless access latency, core network route latency and backhaul latency. Core network route latency mainly depends on the selection of the route path, which can be effectively reduced by software-defined routers. Backhaul latency is mainly affected by the physical distance from the core network to the BS and the physical transmission media. However, it can be significantly reduced and kept at less than 1 ms by using the fiber [1]. Therefore, they're not the focus of our research. Instead, we primarily focus on wireless access latency.

The latency is measured from the moment that the packet is generated until any replica is successfully received and decoded. From the user's perspective, it is influenced by the single access latency and the number of access attempts, i.e., retransmission times. As shown in Fig. 1, the single access latency includes queuing latency, transmission latency, processing latency and propagation latency.

1) Queuing latency: When a newly arrived packet attempts to access for the first time, it must wait for the successful transmission of the already arrived packet. The time consumed by this waiting is called queuing latency, which is denoted as T_q . Queuing latency is usually related to packet generation speed, packet processing speed and queue length.

2) Transmission latency: The time taken from the transmission of the first bit to the last bit of a packet is called transmission latency, which is usually related to the block length of the packet. In general, transmission latency is set to a transmission interval, that is a TTI, denoted as T_t .

3) Processing latency: Processing latency usually refers to the time spent processing the data packet signal, which can be generally divided into receiver processing latency and transmitter processing latency. Receiver processing latency usually includes decoding, demodulation, ACK and no acknowledgment (NACK) signal feedback confirmation and other signal processing latency. Transmitter processing latency refers to the decision whether to perform retransmission or not when the transmitter receives the feedback signal. These two types of latency are collectively referred to as processing latency, which is denoted as T_{proc} . In this paper, we simplify processing latency to about 10 μs [32].

4) Propagation latency: The time it takes for the sent packet to propagate in the wireless channel environment is called propagation latency. This generally depends on signal propagation speed and the distance it travels in the wireless channel. However, in a broad sense, for CF architectures, the backhaul latency of data signal from AP to CPU is also considered part of propagation latency. Propagation latency is denoted as T_{prop} .

Based on above access latency components, we analyze the single access latency. First, the queuing latency is considered. Since mURLLC scenarios are considered, short packet transmission is adopted. The duration of the data phase is expressed as $T_d = L_d/B$. The transmission latency of the overall frame

structure can be denoted as

$$T_t = T_p + T_d. \quad (26)$$

where T_p and T_d are duration of pilot transmission phase and data phase, respectively.

It is worth noting that there are two key reasons why the latency caused by the collision of pilots in data transmission should be included in the access latency. On the one hand, since we adopt the grant-free random access protocol, when a user changes from a silent state to an active state and attempts to access, the user will not require grant and directly send both the preamble and data packets to the AP together. Therefore, if the attempted user fails to access (i.e., collision pilot), the user will retransmit pilot and data packet, resulting in access latency in data packet transmission. On the other hand, the access success probability P_k accounts for pilot collisions. As a result, the expected number of retransmissions inherently accounts for the latency caused by pilot collisions, as pilot collisions reduce the access success probability.

Hence, by considering the number of packets for all active users' buffer as a state variable, the arrival and departure process of packets can be modeled as a M/D/1 queuing model [11]. Specifically, we assume the arrival rate of the data packets is denoted as $\lambda = \sum_{k=1}^K \epsilon_k \lambda_k$, which is related with the activation probability of all users, where ϵ_k denotes the activation probability of users, λ_k denotes the arrival rate of each user. Since the overall service time is T_t , therefore, the average number of data packets in the buffer can be written as $\bar{N} = \frac{\vartheta}{1-\vartheta} (1 - \frac{\vartheta}{2})$, where $\vartheta = \lambda T_t$ denotes a measure of buffer length stability. Thus, based on the Little's formulae [33], the average queuing latency of packets is expressed as

$$T_q = \bar{N} \times T_t = \frac{\vartheta}{1-\vartheta} \left(1 - \frac{\vartheta}{2}\right) \times (T_p + T_d). \quad (27)$$

As shown in Fig. 1, single access latency for the k -th user in CF networks is denoted by T_{ac} , which can be denoted as

$$T_{ac} = T_q + T_t + 2T_{prop} + T_{proc}. \quad (28)$$

Due to retransmissions caused by access failure (i.e., collision, access outage and decoding error) in GFRA, for any attempted access user, it may try to access several times until the user successfully accesses. Let the number of retransmission for the k -th user be R . The overall access latency of the k -th user can be expressed as $T_{oa} = R \times T_{ac}$. It is worth noting that the number of retransmissions for the k -th user is related with the access success probability, which obeys a geometric distribution with parameter $P_{k,a}$ [34]. Therefore, the probability mass function of R_k can be written as

$$f_{R_k}(x) = P_{k,a} (1 - P_{k,a})^{x-1}, \quad x \in \{1, 2, 3, \dots\}. \quad (29)$$

The overall access latency of the k -th user is rewritten as

$$\begin{aligned} T_{oa} &= \mathbb{E}\{R_k\} \times T_{ac} = \sum_{x=1}^{\infty} x f_{R_k}(x) \times T_{ac} \\ &= \sum_{x=1}^{\infty} x (P_{k,a} (1 - P_{k,a})^{x-1}) (T_q + T_t + 3T_{prop} + T_{proc}). \end{aligned} \quad (30)$$

TABLE I
SIMULATION PARAMETERS [27], [28], [33]

Parameters	Value
The number of AP (M)	20, 40, 80
The number of users (K)	50 ~ 400
SNR target	6 dB [28]
The activation probability (ϵ)	0.1
Carrier frequency f_c	1.9 GHz
The maximum transmit power	200 mW
The noise power (σ^2)	-109 dBm
The variance of shadow fading (σ_{sh}^2)	4
Transmission bandwidth (B)	1 MHz
Information size (b)	20 or 32 bytes
Decoding error threshold (ϵ_{th})	1×10^{-5}
Maximum transmission latency threshold (T_t)	1 ms
The block-length (L)	100 ~ 1000
Packet arrival rate (λ_k)	10 pkt/s
Processing latency	10 μ s [32]
Monte-Carlo trials	10^5

IV. SIMULATION RESULTS

In this section, we provide convincing simulation results to verify our proposed massive GFRA in the CF-mMIMO URLLC system. First, we model massive GFRA to analyze the massive access process. Then, we illustrate the key properties (i.e., pilot collision probability, outage probability and decoding error probability) of the user. Finally, we evaluate the access success probability and access latency of the user.

A. Simulation Setting

We consider a CF system of 1 km $\times$ 1 km, which consists of K users, M APs and a CPU. The APs are connected with CPU by the wireless links and collaboratively serve all users. All users are activated by different activation probability, namely, $\Pr(\alpha_k = 1) = \epsilon$ and $\Pr(\alpha_k = 0) = 1 - \epsilon$, respectively, where $\alpha_k \in \{0, 1\}$ is the k -th device activity indicator, if $\alpha_k = 1$, the device is active, otherwise, $\alpha_k = 0$ denotes the device is inactive. The large-scale fading consisting of shadow fading are modeled as β_{mk} [dB] = $-30.5 - 36.7 \log_{10}(d_{mk}/1) + \bar{F}_{mk}$ [28], where $\bar{F}_{mk} \sim \mathcal{N}(0, \sigma_{sh}^2)$ is shadow fading with variance σ_{sh}^2 . d_{mk} denotes the distance from the m -th AP to the k -th user. In the simulation, we set a SNR target that only 95% users can random access this network. More simulation parameters are given in the table I. Furthermore, we select the centralized cellular massive MIMO system [16] and single serving AP case as simulation baseline schemes. For the sake of fairness, we extend [16] to the FBL regime. The centralized cellular massive MIMO system baseline is labeled as ‘‘Co-located mMIMO- [16]’’. The single serving AP case is labeled as ‘‘Single serving AP’’.

B. Simulation Results

First, we demonstrate the decoding error probability for different block-length with 32-byte transmissions at different SINR in Fig. 5. The results show that the decoding error probability decreases gradually as the block-length increases. Similarly, with the increase of SINR, the decoding error probability decreases gradually.

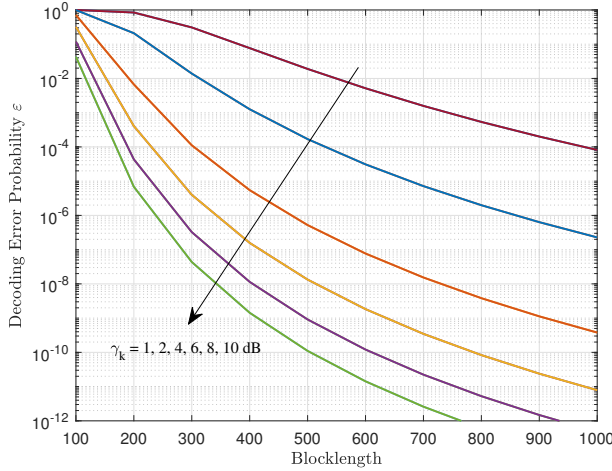


Fig. 5. Decoding error probability for different block-length with 32 byte.

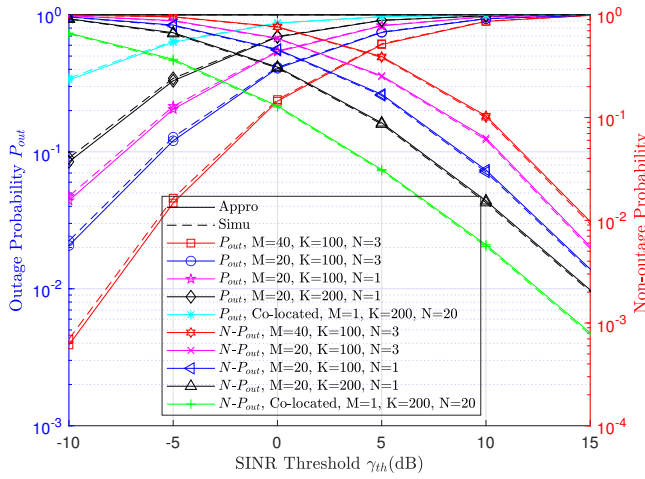


Fig. 6. Performance evaluation of outage probability of the k -th user for different SINR threshold (γ_{th}), different K , AP M and N .

Next, we plot Fig. 6 to show the outage probability of the k -th user for different simulation parameters. The solid and dashed lines represent simulated and approximate values, respectively. The solid and dashed lines represent simulated and approximate values, which are represented as “Simu” and “Appro”, respectively. We select the centralized cellular system as baseline scheme, which is labeled as “Co-located”. From Fig. 6, we observe that the outage probability of the k -th user decreases as the number of APs and antennas increase. With the increase of the number of users, the outage probability decreases. The outage probability increases as the increase of γ_{th} . Furthermore, we can conclude that in the same coverage area, although the number of antennas is the same, the CF-mMIMO system has a higher outage probability compared to the co-located mMIMO architecture. Therefore, the CF-mMIMO architecture has better macro diversity. For non-outage probability, the opposite conclusion can be drawn.

Then, in Fig. 7, we evaluate the access success probability for different numbers of users and different block-length with

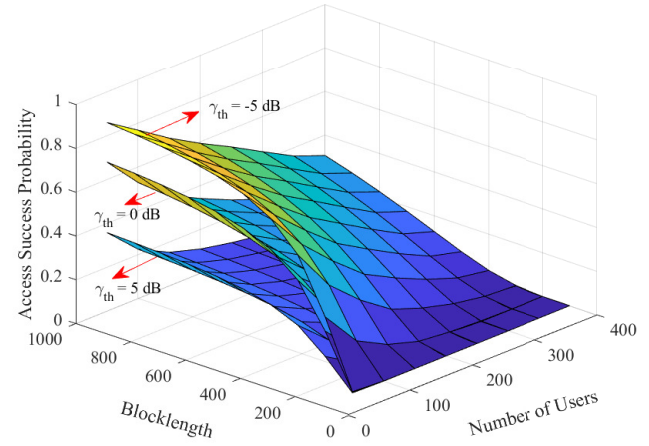


Fig. 7. Access success probability for different number of users vs. block-length, with different $\gamma_{th} = -5, 0, 5$ dB.

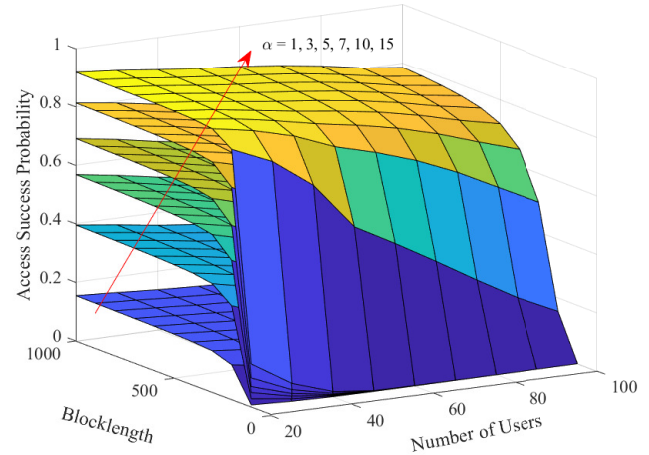


Fig. 8. Performance evaluation of access success probability for multiple APs vs. different number of users and block-length.

given different γ_{th} . From Fig. 7, we observe that the access success probability increases with longer block-length, and decreases as the number of users increases. This can be due to the fact that the decoding error probability decreases as the block-length increases, while the pilot collision probability and the outage probability increases as the number of users increases. These factors affect the access success probability of the user. Furthermore, with the increase of γ_{th} , the access success probability increases gradually. It can be interpreted by the fact that the non-outage probability decreases as γ_{th} increases, which affects the outage probability of the user.

The performance evaluation of access success probability for the multiple APs scheme is given in Fig. 8. First, similar conclusions can be drawn as Fig. 7. However, the access success probability enhances with the increase of α . It is worth noting that α is limited due to practical deployment and the user centric-based property of the CF system. However, we can further improve the access success probability by increasing

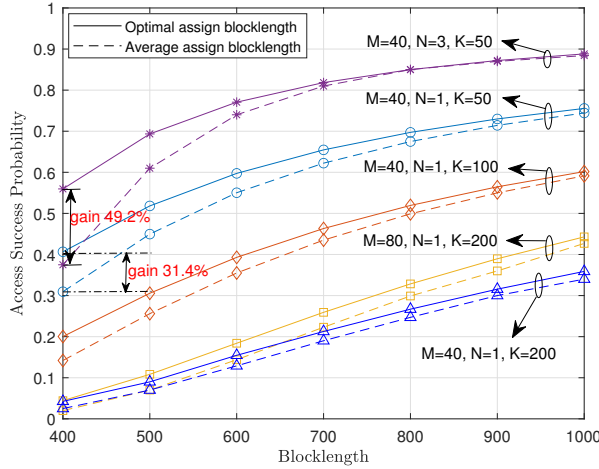


Fig. 9. Access performance evaluation of joint optimization scheme for different block-length, different parameters with $\gamma_{th} = -5$ dB.

the number of antennas at each AP.

Fig. 9 demonstrates the access performance of the joint optimization scheme for given different total block-length and different system parameters, which is denoted as “optimal assign blocklength”. We choose the average assign block-length as a comparison scheme, which is represented as “Average assign blocklength”. From Fig. 9, we observe that whether “Optimal assign block-length” or “Average assign blocklength”, the access reliability increases as block-length increases. Then, with the increase of AP and antenna, the access success probability improves gradually when the block-length and the number of users are given. This can be interpreted by the fact that decoding error probability improves as block-length increases, and non-outage probability can be improved with more antennas. There is even 49.2% performance gain when the block-length is $L = 400$, $M = 40$, $N = 3$ and $K = 50$. Similarly, compared to “average assign block-length”, there is a 31.4% performance gain when $L = 400$, $M = 40$, $N = 1$ and $K = 50$. However, the performance gain is still small as block-length increases when $M = 40$, $N = 1$ and $K = 200$. This can be interpreted by the fact that too many users exceed the access capacity of the network, causing a high outage probability, which is easy to trigger access failures and leads to poor performance gains.

In Fig. 10, we provide the single access latency evaluation for different block-length and arrival rates. With the increase of the block-length, the access latency gradually increases, which can be attributed to the increase in transmission latency. Then, access latency increases with a higher arrival rate when the block-length is given. It can be interpreted by that the increase of arrival rate results in a corresponding rise in queuing latency.

In Fig. 11, we evaluate access performance of the user with various parameters, and compare various baseline schemes. From Fig. 11, we can observe that in general, the access performance of CF-mMIMO system outperforms that of the massive MIMO scenario under identical conditions, and the single serving AP case shows the worst access performance.

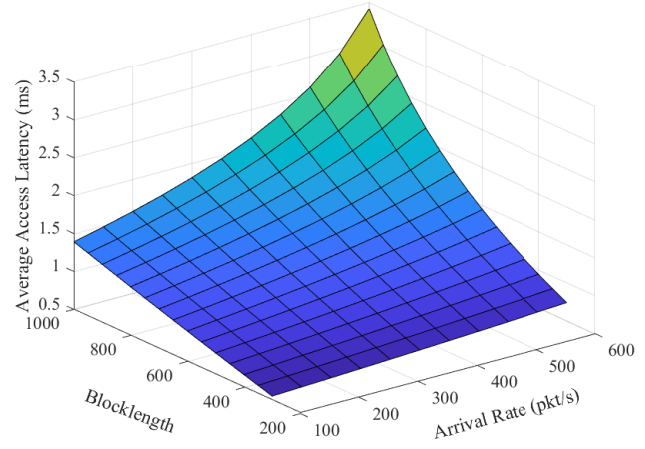


Fig. 10. Single access latency for different arrival rate λ vs. block-length

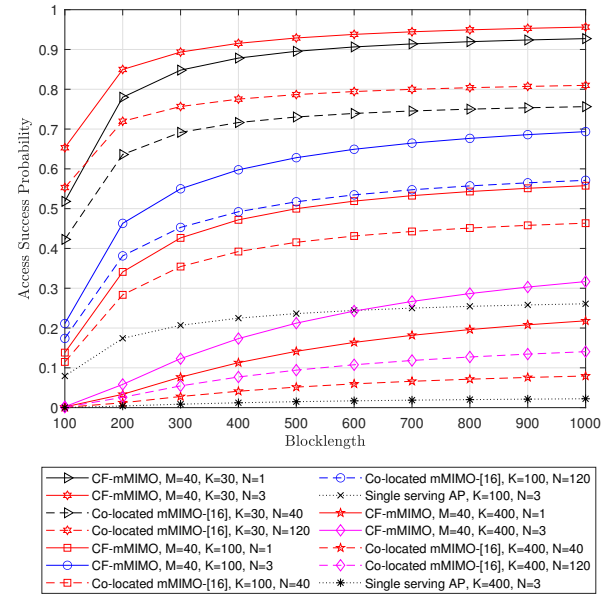


Fig. 11. Performance evaluation for access success probability of the k -th user vs. blocklength with different K , M and N , with $\gamma_{th} = 5$ dB.

This is because the CF-mMIMO system offers superior macro diversity in massive random access. As the block-length increases, the access success probability of the user gradually improves. This is attributed to the reduction in decoding error probability associated with longer blocklengths in the FBL regime. Moreover, as the number of users increases, the access success probability decreases across all schemes. In particular, in the single AP case, a significant user load can drastically reduce the access success probability, likely due to the system exceeding the serving capacity of the BS (AP).

Fig. 12 demonstrates the access success probability of users for different block-length, with other parameters, such as K , M , N and $\gamma_{th} = 5$ dB. From this figure, in general, the access success probability improves with the increase of block-length. Then, as the number of APs and antennas increases,

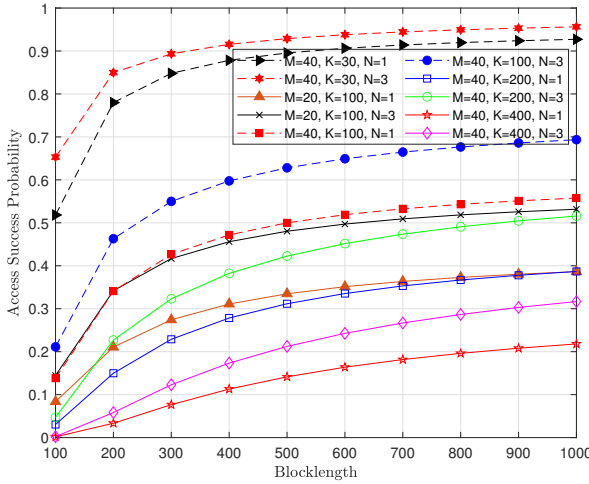


Fig. 12. Performance evaluation of access success probability of the k -th user for different block-length, different K , M and N with $\gamma_{th} = 5$ dB.

the access success probability gradually enhances. However, the access success probability decreases as the number of users increases.

Additionally, in Fig. 13, we evaluate the relationship between block-length and the number of retransmissions, with given M , N and K . From this figure, we observe that the number of retransmissions gradually decreases as the block-length increase, which may be due to the improvement in the access success probability with increased block-length.

Furthermore, based on single access latency and the number of retransmissions, we further evaluate the total access latency of users for different block-length under the given M , N , K and arrival rate λ . In Fig. 14, the access latency performance is evaluated with different block-lengths and various packet arrival rates. From Fig. 14, the total access latency initially decreases and then increases as block-length increases when $\lambda = 400$ pkt/s. This can be interpreted by the fact that when the packet arrival rate is high, the queuing latency dominates the total access latency. The initial increase of the block length causes a reduction in the access success probability, resulting in fewer retransmissions and lower queuing latency, thereby reducing the total access latency. However, as the block-length further increases, the gain in access success probability is minimal, the transmission latency gradually dominates the total access latency, resulting in gradually a gradual increase in transmission latency as the block-length increases. Whereas, at smaller arrival rates, transmission latency remains the primary factor, increasing gradually with the increase of block-length.

V. CONCLUSION

In this paper, we have studied massive GFRA in the CF-mMIMO URLLC system, and further analyzed the access performance. First, we gave a massive GFRA model based on the CF-mMIMO URLLC system. We then derived the SINR expression for the active user and approximated it as

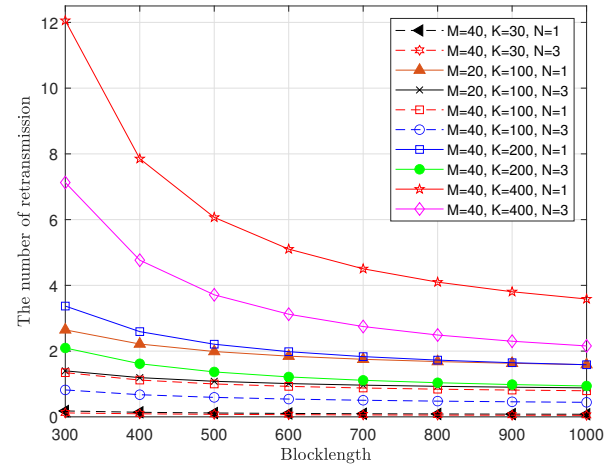


Fig. 13. Performance evaluation for retransmissions number vs. blocklength.

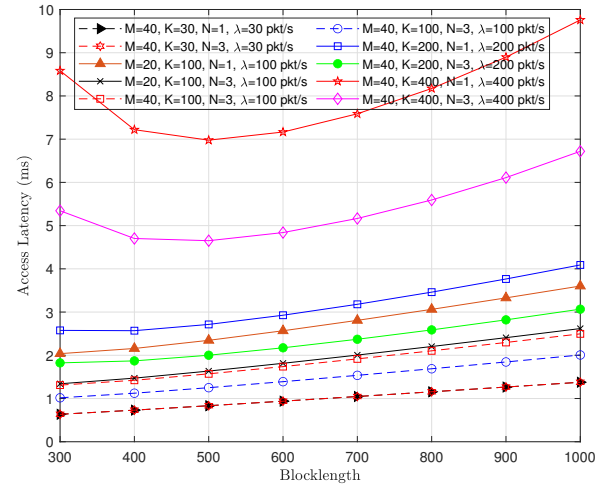


Fig. 14. Performance evaluation for access latency vs. blocklength with different λ .

Log-normal distribution by leveraging the properties of CF-mMIMO. Based on this, we derived the expression for the access success probability of the attempted access users to measure access reliability by comprehensively considering the collision probability, the decoding error probability and outage probability. Furthermore, leveraging the macro diversity of the CF-mMIMO system, we proposed a scalable user-centric-based multiple APs access scheme to enhance access reliability. On the other hand, we analyzed the components of the access latency based on access procedure and associated GFRA protocol. Subsequently, the access latency was analyzed by further incorporating access reliability and the number of retransmissions. Finally, simulation results demonstrated the validity and effectiveness of the access reliability and access latency analysis in the CF-mMIMO URLLC system.

APPENDIX A THE PROOF OF THEOREM 1

We begin to analysis the conclusion of Theorem 1. Due to $\Phi_k \sim \mathcal{LN}(\mu_{\Phi_k}, \sigma_{\Phi_k}^2)$ and $\Psi_k \sim \mathcal{LN}(\mu_{\Psi_k}, \sigma_{\Psi_k}^2)$, we can know $\log(\Phi_k) \sim \mathcal{N}(\mu_{\Phi_k}, \sigma_{\Phi_k}^2)$ and $\log(\Psi_k) \sim \mathcal{N}(\mu_{\Psi_k}, \sigma_{\Psi_k}^2)$, whose mean and variance can be computed as

$$\mu_{\Phi_k} = \log\left(\frac{\mathbb{E}[\Phi_k]^2}{\sqrt{\mathbb{E}[\Phi_k^2]}}\right), \sigma_{\Phi_k}^2 = \log\left(\frac{\mathbb{E}[\Phi_k^2]}{(\mathbb{E}[\Phi_k])^2}\right) \quad (\text{A.1})$$

and

$$\mu_{\Psi_k} = \log\left(\frac{\mathbb{E}[\Psi_k]^2}{\sqrt{\mathbb{E}[\Psi_k^2]}}\right), \sigma_{\Psi_k}^2 = \log\left(\frac{\mathbb{E}[\Psi_k^2]}{(\mathbb{E}[\Psi_k])^2}\right) \quad (\text{A.2})$$

where the expectation term $\mathbb{E}[\Phi_k]$, $\mathbb{E}[\Phi_k^2]$, $\mathbb{E}[\Psi_k]$ and $\mathbb{E}[\Psi_k^2]$ can refer to the Appendix. B in [35].

And then, due to $\gamma_k = \frac{\Phi_k}{\Psi_k}$, after logarithmic operation for the two sides of above equation, we can obtain that $\log(\gamma_k) = \log\left(\frac{\Phi_k}{\Psi_k}\right) = \log(\Phi_k) - \log(\Psi_k)$. Since $\log(\Phi_k)$ and $\log(\Psi_k)$ all follow the normal distribution. So $\log(\gamma_k)$ also yields the normal distribution. Namely, $\log(\gamma_k) \sim \mathcal{N}(\mu_{\gamma_k}, \sigma_{\gamma_k}^2)$. According to the property of normal distribution, the mean μ_{γ_k} can be expressed as

$$\mu_{\gamma_k} = \mathbb{E}[\log(\Phi_k)] - \mathbb{E}[\log(\Psi_k)] = \mu_{\Phi_k} - \mu_{\Psi_k}. \quad (\text{A.3})$$

The variance $\sigma_{\gamma_k}^2$ can be computed as

$$\begin{aligned} \sigma_{\gamma_k}^2 &= \mathbb{V}\left(\log(\Phi_k) - \log(\Psi_k)\right) \\ &= \sigma_{\Phi_k}^2 + \sigma_{\Psi_k}^2 - 2\text{Cov}\left(\log(\Phi_k), \log(\Psi_k)\right) \\ &= \sigma_{\Phi_k}^2 + \sigma_{\Psi_k}^2 - 2\log\left(\frac{\mathbb{E}[\Phi_k \Psi_k]}{\mathbb{E}[\Phi_k] \mathbb{E}[\Psi_k]}\right), \end{aligned} \quad (\text{A.4})$$

where $\text{Cov}(\log(\Phi_k), \log(\Psi_k))$ is covariance of $\log(\Phi_k)$ and $\log(\Psi_k)$. The expectation term $\mathbb{E}[\Phi_k \Psi_k]$ is omitted and can be calculated as Appendix B in [35]. This completes the proof.

APPENDIX B THE PROOF OF THEOREM 2

Since γ_k follows log-normal distribution, so we can know that the PDF of γ_k can be expressed as

$$f(x) = \frac{1}{\sqrt{2\pi x \sigma_{\gamma_k}}} \exp\left(-\frac{(\ln x - \mu_{\gamma_k})^2}{2\sigma_{\gamma_k}^2}\right). \quad (\text{B.1})$$

Thus, we can get

$$\begin{aligned} \mathbb{P}(\gamma_k \geq \gamma_{th}) &= 1 - \mathbb{P}(\gamma_k < \gamma_{th}) = 1 - \int_0^{\gamma_{th}} f(x) dx \\ &= 1 - \int_0^{\gamma_{th}} \frac{1}{\sqrt{2\pi x \sigma_{\gamma_k}}} \exp\left(-\frac{(\ln x - \mu_{\gamma_k})^2}{2\sigma_{\gamma_k}^2}\right) dx. \end{aligned} \quad (\text{B.2})$$

Then, we perform the substitution operation for the above integral operation. Let $\frac{\ln x - \mu_{\gamma_k}}{\sqrt{2\sigma_{\gamma_k}^2}} = \omega$. So, $\ln x = \sqrt{2}\omega\sigma_{\gamma_k} + \mu_{\gamma_k}$. And $dx = \sqrt{2}x\sigma_{\gamma_k} d\omega$. For this, we can further calculate the values of the upper and lower limits of the integral as

$$\omega = \begin{cases} \frac{\ln \gamma_{th} - \mu_{\gamma_k}}{\sqrt{2}\sigma_{\gamma_k}}, & x = \gamma_{th} \\ -\infty, & x = 0 \end{cases}, \quad (\text{B.3})$$

Therefore, Eq. (B.2) can be represented as

$$\begin{aligned} \mathbb{P}(\gamma_k \geq \gamma_{th}) &= 1 - \int_0^{\gamma_{th}} \frac{1}{\sqrt{2\pi x \sigma_{\gamma_k}}} e^{-\omega^2} dx \\ &= 1 - \int_{-\infty}^{\frac{\ln \gamma_{th} - \mu_{\gamma_k}}{\sqrt{2}\sigma_{\gamma_k}}} \frac{1}{\sqrt{2\pi x \sigma_{\gamma_k}}} e^{-\omega^2} \sqrt{2}x\sigma_{\gamma_k} d\omega \\ &= 1 - \int_{-\infty}^{\frac{\ln \gamma_{th} - \mu_{\gamma_k}}{\sqrt{2}\sigma_{\gamma_k}}} \frac{1}{\sqrt{\pi}} e^{-\omega^2} d\omega \\ &\stackrel{(a)}{=} 1 - \frac{1}{2} \text{erfc}\left(-\frac{\ln(\gamma) - \mu_{\gamma_k}}{\sqrt{2}\sigma_{\gamma_k}}\right), \end{aligned} \quad (\text{B.4})$$

where the step (a) holds due to the definition of the complementary error function, i.e., $\text{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^\infty e^{-t^2} dt$. This completes the proof.

APPENDIX C THE PROOF OF EQ. (18)

Since we introduced a Q function approximation, namely, $Q\left(\sqrt{\frac{L_d}{V(\gamma_k)}} \left[\log_2(1 + \gamma_k) - \frac{b}{L_d}\right]\right) \approx \varphi(\gamma_k)$, and then, we can obtain that

$$\mathbb{E}\left\{Q\left(\sqrt{\frac{L_d}{V(\gamma_k)}} \left[\log_2(1 + \gamma_k) - \frac{b}{L_d}\right]\right)\right\} \approx \mathbb{E}\{\varphi(\gamma_k)\}. \quad (\text{C.1})$$

Based on Eq. (11), we further derived $\mathbb{E}\{\varphi(\gamma_k)\}$ as follow,

$$\begin{aligned} \mathbb{E}\{\varphi(\gamma_k)\} &= \int_0^{\phi_1} f_{\gamma_k}(s) ds + \int_{\phi_1}^{\phi_2} \left[\frac{1}{2} - \psi\sqrt{L_d}(s - \theta)\right] f_{\gamma_k}(s) ds \\ &= F_{\gamma_k}(s) \Big|_0^{\phi_1} + \left(\frac{1}{2} + \psi\sqrt{L_d}\theta\right) F_{\gamma_k}(s) \Big|_{\phi_1}^{\phi_2} - \psi\sqrt{L_d} \int_{\phi_1}^{\phi_2} s f_{\gamma_k}(s) ds \\ &= F_{\gamma_k}(\phi_1) - F_{\gamma_k}(0) + \left(\frac{1}{2} + \psi\sqrt{L_d}\theta\right) (F_{\gamma_k}(\phi_2) - F_{\gamma_k}(\phi_1)) \\ &\quad - \psi\sqrt{L_d} \int_{\phi_1}^{\phi_2} s f_{\gamma_k}(s) ds \\ &\stackrel{(a)}{=} F_{\gamma_k}(\phi_1) + \left(\frac{1}{2} + \psi\sqrt{L_d}\theta\right) (F_{\gamma_k}(\phi_2) - F_{\gamma_k}(\phi_1)) \\ &\quad - \psi\sqrt{L_d} \left\{ \left[s F_{\gamma_k}(s) \right] \Big|_{\phi_1}^{\phi_2} - \int_{\phi_1}^{\phi_2} F_{\gamma_k}(s) ds \right\} \\ &= F_{\gamma_k}(\phi_1) + \left(\frac{1}{2} + \psi\sqrt{L_d}\theta\right) (F_{\gamma_k}(\phi_2) - F_{\gamma_k}(\phi_1)) \\ &\quad - \psi\sqrt{L_d} \left\{ \phi_2 F_{\gamma_k}(\phi_2) - \phi_1 F_{\gamma_k}(\phi_1) - \int_{\phi_1}^{\phi_2} F_{\gamma_k}(s) ds \right\} \\ &= \left(\frac{1}{2} + \psi\sqrt{L_d}(\theta - \phi_2)\right) F_{\gamma_k}(\phi_2) - \left(\psi\sqrt{L_d}(\theta - \phi_1) - \frac{1}{2}\right) F_{\gamma_k}(\phi_1) \\ &\quad + \psi\sqrt{L_d} \int_{\phi_1}^{\phi_2} F_{\gamma_k}(s) ds \stackrel{(b)}{=} \psi\sqrt{L_d} \int_{\phi_1}^{\phi_2} F_{\gamma_k}(s) ds \end{aligned} \quad (\text{C.2})$$

where $F_{\gamma_k}(s)$ denotes the CDF of γ_k . And the step (a) holds because of $F_{\gamma_k}(0) = 0$. The last step (b) holds due to $\phi_1 = \theta - \frac{1}{2\psi\sqrt{L_d}}$ and $\phi_2 = \theta + \frac{1}{2\psi\sqrt{L_d}}$. This finishes the proof.

APPENDIX D THE PROOF OF THEOREM 3

To simply $\int_{\phi_1}^{\phi_2} F_{\gamma_k}(s) ds$, we first introduce the CDF of γ_k . In this paper, we consider CF-mMIMO scenarios. By leveraging the channel hardening character of CF-mMIMO [8], the expression of γ_k can be represented as

$$\gamma_k = \frac{|\hat{\mathbf{h}}_k|^2}{\sum_{i \neq k}^K \frac{|\hat{\mathbf{h}}_k \hat{\mathbf{h}}_i|^2}{|\hat{\mathbf{h}}_k|^2} + \text{Tr}(\mathbf{C}_{\hat{\mathbf{h}}_k}) + \frac{\sigma^2}{\rho_d}} = \frac{A}{B + D}. \quad (\text{D.1})$$

where $A = |\hat{\mathbf{h}}_k|^2$, which is the sum of MN complex Gaussian variables square. Therefore, $A \sim \Gamma(MN, 1)$. The PDF of A can be expressed as $f_A(x) = \frac{x^{MN-1}}{(MN-1)!} e^{-x}$. $B = \sum_{i \neq k}^K \frac{|\hat{\mathbf{h}}_k \hat{\mathbf{h}}_i|^2}{|\hat{\mathbf{h}}_k|^2}$ is the sum of $K-1$ independent exponential random variables, which is equivalent to following $\Gamma(K-1, 1)$ [36]. Therefore, $B \sim \Gamma(K-1, 1)$, whose probability density function is $f_B(b) = \frac{b^{K-2}}{(K-2)!} e^{-b}$. Since $D = \text{Tr}(\mathbf{C}_{\hat{\mathbf{h}}_k}) + \frac{\sigma^2}{\rho_d}$ is a constant. Let $C = B + D$, the denominator of γ_k yields multiple parameters Γ distribution with $C \sim \Gamma(K-1, 1, D)$, whose PDF can be expressed as $f_C(c) = \frac{(c-D)^{K-2}}{(K-2)!} e^{-(c-D)}$. Therefore, we can derive

$$f_{\gamma_k}(x, c) = f_A(x) f_C(c) = \frac{x^{MN-1} (c-D)^{K-2}}{(MN-1)! (K-2)!} e^{-(x+c-D)}. \quad (\text{D.2})$$

for $x > 0$ and $c > D$, by applying the variable substitution, we can get that

$$\begin{aligned} f_{\gamma_k}(\gamma, c) &= \frac{\partial(\gamma c)}{\partial \gamma} f_{\gamma_k}(\gamma c, c) \\ &= \frac{\gamma^{MN-1} c^N (c-D)^{K-2}}{(MN-1)! (K-2)!} e^{-(\gamma+1)c} e^D. \end{aligned} \quad (\text{D.3})$$

Further, eliminate the variable C , we can get that

$$\begin{aligned} f_{\gamma_k}(\gamma) &= \int_{-\infty}^{+\infty} f_{\gamma_k}(\gamma, c) dc = \frac{\gamma^{MN-1} e^D}{(MN-1)! (K-2)!} \\ &\times \int_D^{+\infty} D^{MN} e^{-(\gamma+1)c} (c-D)^{K-2} dc. \end{aligned} \quad (\text{D.4})$$

Using the Whittaker function [37] and Laguerre polynomials [38], we can further obtain that

$$f_{\gamma_k}(\gamma) = (-1)^{MN} \frac{MN \gamma^{MN-1} L_{MN}^{(1-K-MN)}(D(1+\gamma))}{(1+\gamma)^{MN+K-1}} e^{-D\gamma}. \quad (\text{D.5})$$

where $L_n^\alpha(x)$ denotes the Laguerre polynomials.

In considered CF-mMIMO systems, since the total number of antennas exceeds active users, namely, $MN > K \gg K_a$. Additionally, for small value D , the argument of Laguerre polynomial can be approximated by its high order term. In other words, for small value γ , the PDF of γ_k is not zero. Therefore, (D.5) can be approximated as

$$f_{\gamma_k}(\gamma) \approx \frac{\gamma^{MN-1} D^{MN}}{(MN-1)!} e^{-D\gamma}. \quad (\text{D.6})$$

Based on this, we can further derive that $F_{\gamma_k}(s) = \mathbb{P}\{\gamma \leq s\}$. we can further denote the CDF of SINR as [39]

$$\begin{aligned} F_{\gamma_k}(s) &= \mathbb{P}\{\gamma \leq s\} = \int_0^s \frac{\gamma^{MN-1} D^{MN}}{(MN-1)!} e^{-D\gamma} d\gamma \\ &\stackrel{x=D\gamma, z=Ds}{=} \int_0^z \frac{x^{MN-1}}{(MN-1)!} e^{-x} dx \\ &= -\frac{x^{MN-1}}{(MN-1)!} e^{-x} \Big|_0^z + \int_0^z \frac{x^{MN-2}}{(MN-2)!} e^{-x} dx. \end{aligned} \quad (\text{D.7})$$

And then, let $\Upsilon_n(z) = \int_0^z \frac{x^n}{n!} e^{-x} dx$, so that the CDF of SINR can be represented as $F_{\gamma_k}(s) = \Upsilon_{MN-1}(z) = -\frac{z^{MN-1}}{(MN-1)!} e^{-z} + \Upsilon_{MN-2}(z)$. Due to $\Upsilon_0(z) = 1 - e^{-z}$, we can obtain that

$$\int_0^z \frac{x^{MN-1}}{(MN-1)!} e^{-x} dx = \Upsilon_{MN-1}(z) = 1 - e^{-z} \sum_{n=0}^{MN-1} \frac{z^n}{n!}. \quad (\text{D.8})$$

Based on the last equation, we further obtain that

$$\begin{aligned} \int_{\phi_1}^{\phi_2} F_{\gamma_k}(s) ds &= \int_{\phi_1}^{\phi_2} 1 - e^{-z} \sum_{n=0}^{MN-1} \frac{z^n}{n!} dz \\ &= \frac{1}{D} \int_{u_1}^{u_2} \left[1 - e^{-z} \sum_{n=0}^{MN-1} \frac{z^n}{n!} \right] dz \\ &= \frac{1}{D} \left[(u_2 - u_1) - \sum_{n=0}^{MN-1} \left(\int_0^{u_2} \frac{z^n e^{-z}}{n!} dz - \int_0^{u_1} \frac{z^n e^{-z}}{n!} dz \right) \right]. \end{aligned} \quad (\text{D.9})$$

where $u_1 = D\phi_1$, $u_2 = D\phi_2$.

From (D.8), we can know that (D.9) can be rewritten as

$$\begin{aligned} \int_{\phi_1}^{\phi_2} F_{\gamma_k}(s) ds &= \frac{1}{D} \left[(u_2 - u_1) - \sum_{n=0}^{MN-1} \left(e^{-u_1} \frac{u_1^n}{n!} - e^{-u_2} \frac{u_2^n}{n!} \right) \right] \\ &= \frac{1}{D} \left[(u_1 - u_2) - \sum_{n=0}^{MN-1} (MN - n) \tilde{\Upsilon}_n \right]. \end{aligned} \quad (\text{D.10})$$

where $\tilde{\Upsilon}_n = \frac{u_1^n}{n!} e^{-u_1} - \frac{u_2^n}{n!} e^{-u_2}$. By substituting Eq. (D.10) into (18), we can obtain (19). This completes the proof.

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